

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Annexure A

Duration : 1 Hour
Total Marks:50

Design Task

Code of Conduct:

I Kishore S.P (192511081) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Design Task

- 1. Hybrid AI Recommendation Engine Design**
 Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.
- 2. Smart Disaster Detection System**
 Develop a distributed AI system using PySpark and RDD operations to detect natural disasters (such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.
- 3. AI-Based Fraud Detection Framework**
 Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.
- 4. Autonomous Supply Chain Optimization System**
 Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory, logistics, demand forecasting). Describe how RDD transformations support large-scale data processing and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.
- 5. Personalized Learning Analytics Platform**
 Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

1. Hybrid AI-Based Recommendation System

A hybrid recommendation system combines content-based filtering and collaborative filtering.
Working

- Content-based filtering recommends items similar to those the user previously liked.
- Collaborative filtering recommends items based on preferences of similar users.
- PySpark RDDs distribute large user-item datasets across multiple machines.

RDD Processing

For example, RDDs can group users by product ratings and calculate similarity between users or items.

Intelligent Agent

The recommendation agent:

- Monitors clicks, purchases, ratings, and searches.
- Updates the user's preference profile.
- Combines content and collaborative scores.
- Continuously adapts recommendations based on feedback.

Result: Personalized recommendations that can scale to millions of users and products.

2. Smart Disaster Detection System

The system combines data from satellites, IoT sensors, weather stations, and social media to detect disasters such as floods and earthquakes.

Architecture

RDD Operations

- `map()` → extract useful features
- `filter()` → remove irrelevant/noisy data
- `groupByKey()` → group information by location/time
- `reduceByKey()` → aggregate sensor measurements

Data Fusion

Information from different sources is combined to improve confidence.

For example:

Intelligent Agents

- Sensor agents: collect environmental information.
- Analysis agents: detect abnormal patterns.
- Decision agents: estimate disaster severity.
- Communication agents: issue warnings and coordinate response.

Result: Faster disaster detection and early warning.

3. AI-Based Fraud Detection Framework

The system analyzes large volumes of financial transactions and identifies suspicious activities.

Pipeline

RDD Processing

Transaction records can be distributed across nodes.

Typical operations:

- `map()` → calculate transaction features
- `filter()` → remove invalid transactions
- `groupByKey()` → group transactions by customer/account
- `reduceByKey()` → calculate totals and statistics

Anomaly Detection

The system can identify unusual:

- Transaction amounts
- Locations
- Transaction frequency
- Login behavior
- Spending patterns

Methods such as GMM, clustering, statistical thresholds, or machine-learning classifiers can be used.

Intelligent Agents

- Monitoring agent: observes transactions.
- Fraud detection agent: identifies suspicious behavior.
- Alert agent: generates warnings.
- Learning agent: learns from confirmed fraud and legitimate transactions.

Result: Scalable and continuously improving fraud detection.

4. Autonomous Supply Chain Optimization System

The system uses AI and PySpark to optimize:

- Inventory
- Transportation
- Warehouses
- Demand forecasting
- Delivery routes

Architecture

RDD Transformations

- `map()` → calculate demand and inventory features

- filter() → remove invalid records
- groupByKey() → group data by warehouse/product
- reduceByKey() → calculate total demand and inventory
- sortByKey() → rank products or routes

Intelligent Agents

- Inventory agent: determines when stock should be replenished.
- Warehouse agent: manages warehouse resources.
- Logistics agent: selects efficient transportation options.
- Demand agent: forecasts future demand.

Agents communicate and make decentralized decisions.

Goal:

while maintaining sufficient inventory.

5. Personalized Learning Analytics Platform

This platform analyzes student behavior and automatically provides personalized learning content.

Input Data

- Quiz scores
- Assignment results
- Video viewing time
- Course progress
- Student interactions
- Learning history

Architecture

RDD Operations

- map() → calculate student performance features
- filter() → remove incomplete records
- groupByKey() → group activities by student
- reduceByKey() → calculate performance statistics

Intelligent Learning Agent

The agent monitors each learner and:

- Detects weak topics.
- Adjusts content difficulty.
- Recommends additional exercises.
- Provides immediate feedback.
- Increases difficulty when performance improves.

For example:

Result

The system can support thousands of learners simultaneously while providing scalable, adaptive, and personalized education.

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	Data Flow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete
Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations/actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration
Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed
Data Flow & Processing Pipeline	Clear, logical, and well-defined data flow	Mostly clear	Some gaps	Poorly defined

Innovation & Creativity	Highly innovative, realistic, and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation & Documentation	Very clear, neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poor presentation